



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NBA for UG Programs

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

2023-2024



PUBLISHED BY;
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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ABOUT THE DEPARTMENT

It is a pleasure to Head of Artificial Intelligence and Data Science(AD) at Ramco Institute of Technology, Rajapalayam. The department has a team of highly experienced and motivated faculty members. All the faculty members are well qualified and competitive enough in preparing the young mind (our students) for global completion.

Students of AD department are highly motivated and ready for IT Industry with hands on experience on cutting edge technologies and programming languages. I assure that students will achieve university rank positions and will be placed with high paid jobs. Students are facilitated with placement training in both aptitude and technical sessions, communication classes for better placement.

Sports, Co-Curricular and Extra-curricular activities take place at institute level, Intra college and inter college level to improve their professional development. Departments works for overall growth of students and inculcate the qualities that are required and acceptable by society.

VISION

To impart international quality education, promote collaborative research and graduate industry-ready engineers in the domain of Artificial Intelligence and Data Science to serve the society.

MISSION

Excel in Teaching-Learning process and collaborative Research by the use of modern infrastructure and innovative components.

Establish an Artificial Intelligence and Data Science based centre of excellence to prepare professional technocrats for solving interdisciplinary industry problems in various applications

Motivate students to emerge as entrepreneurs with leadership qualities in a societal centric programme to fulfil Industry and community needs with ethical standards

PEO'S

After successful completion of the degree, the students will be able to

PEO 1. Apply Artificial Intelligence and Data Science techniques with industrial standards and pioneering research to solve social and environment-related problems for making a sustainable ecosystems.

PEO 2. Excel with professional skills, fundamental knowledge, and advanced futuristic technologies to become Data Scientists, Data Analyst Managers, Data Science leaders AI Research Scientists, or Entrepreneurs.

PO's

Engineering Graduates will be able to:

Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem Analysis

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics :

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and

finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning:

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO

After successful completion of the degree, the students will be able to:

PSO 1: To apply analytic technologies to arrive at actionable foresight, Insight, hindsight from data for solving business and engineering problems.

PSO 2: To create, and apply the techniques of AI and Data Science to forecast future events in the domain of Healthcare, Education, and Agriculture, Manufacturing, Automation, Robotics, Transport, etc.

PSO 3: To enrich the critical thinking skills in emerging technologies such as Hybrid Mobile application development, cloud technology stack, and cyber-physical systems with mathematical aid to foresee the research findings and provide the solutions.

PYTHON ROADMAP

BASICS

- Basics Syntax
- Variable
- Data Types
- Conditionals
- Typecasting
- Exceptions
- Functions
- List, Tuples, Sets
- Dictionaries

ADVANCE

- List
- Comprehensions
- Generators
- Expressions
- Paradigms
- Regex
- Decorators
- Iterators
- Lambdas

DSA

- Arrays and
- Linked Lists
- Heaps, Stacks,
- Queues
- Hash Tables
- Binary Search
- Trees
- Recursion, Sorting
- Algorithms

OOP

- Classes
- Inheritance
- Methods
(Including
Dunder
Methods)

DATA SCIENCE

- Numpy, Pandas
- Matplotlib
- Seaborn
- Scikit-Learn
- Tensorflow
- Pytorch

PACKAGE MANAGERS

- PyPI
- pip
- conda

WEB FRAMEWORKS

- Django
- Flask
- FastAPI
- Tornado

AUTOMATION

- File Manipulation
[OS, shutil, pathlib]
- Web Scraping
[BeautifulSoup, Scrapy]
- GUI Automation
[PyAutoGUI]
- Network Automation

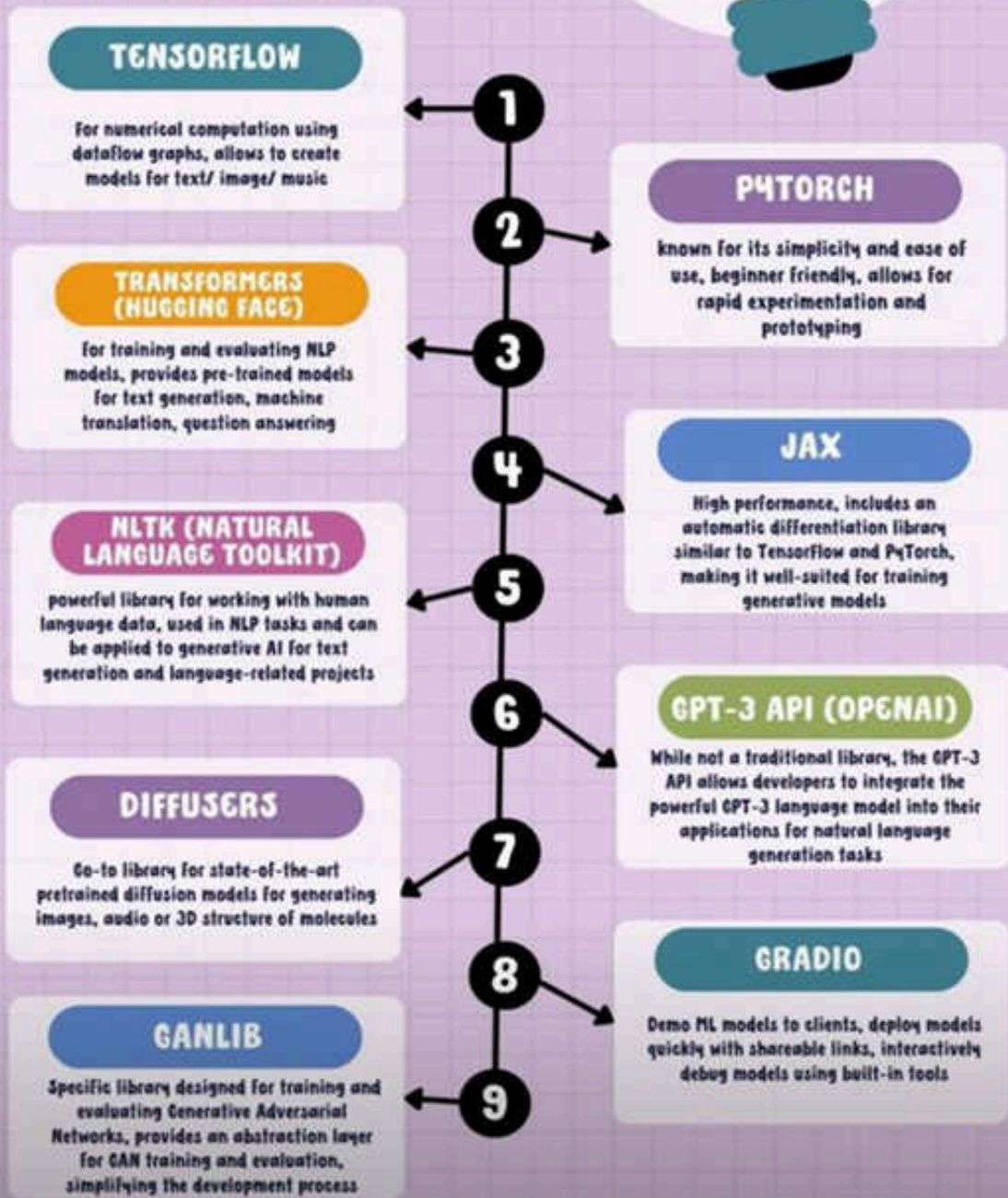
TESTING

- Unit Testing
[unittest, pytest]
- Integration Testing
- End-to-End Testing
[Selenium, PyAutoGUI]
- Test-Driven
Development(TDD)

DR.M.KALIAPPAN
PROFESSOR & HEAD
AI & DS

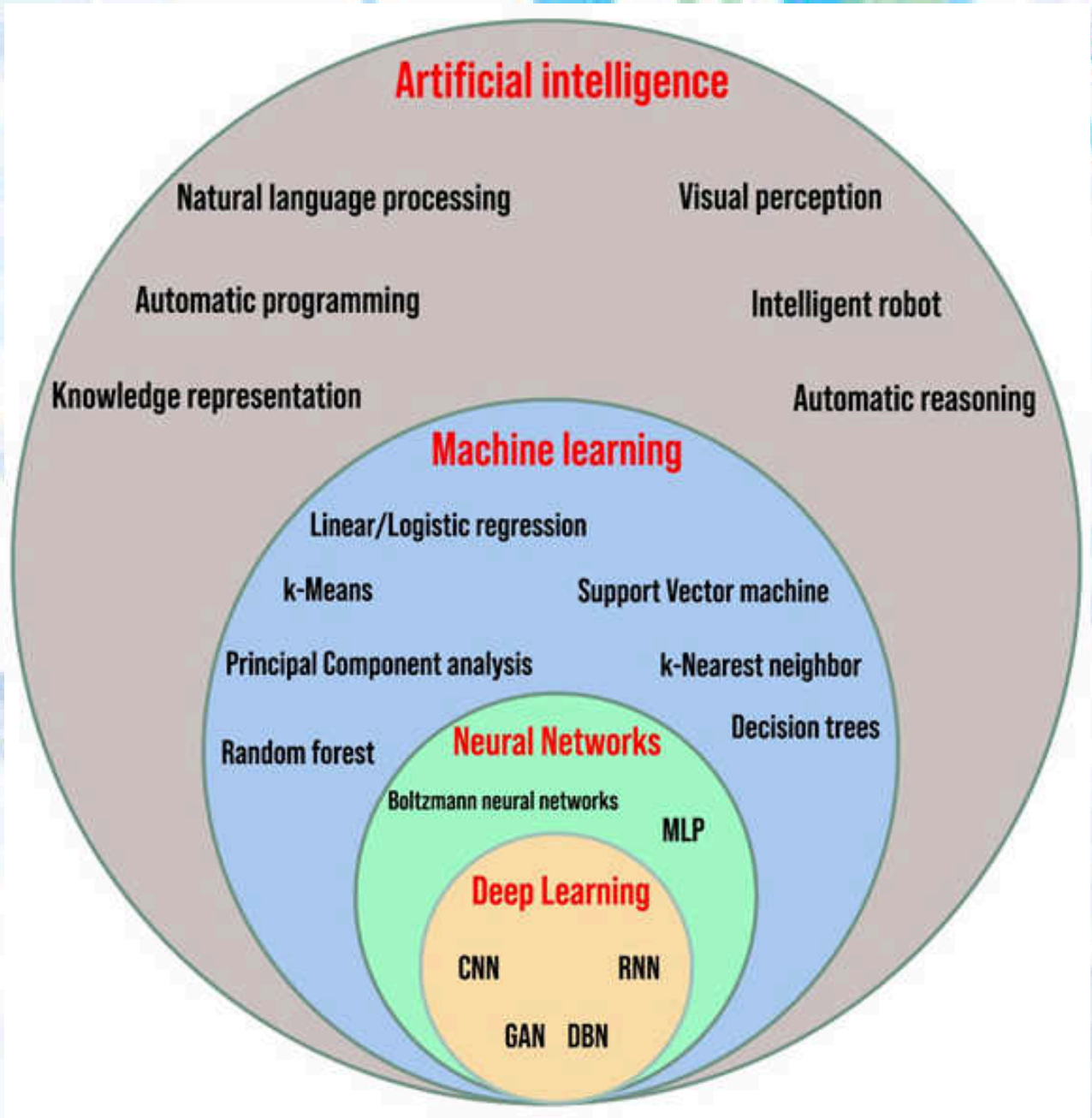
PYTHON LIBRARIES

GENERATIVE AI



DR.M.KALIAPPAN
PROFESSOR & HEAD
AI & DS

RELATIONSHIP BETWEEN AI, ML, NEURAL NETWORKS AND DEEP LEARNING



DR.M.KALIAPPAN
PROFESSOR & HEAD
AI & DS

ARTIFICIAL INTELLIGENCE IMPACT ON EMPLOYMENT

Automation:

AI is increasingly being used to automate repetitive and routine tasks, such as data entry and analysis, customer service, and manufacturing. This can lead to increased efficiency and productivity, but it also has the potential to eliminate some jobs.

New job opportunities:

While some jobs may be eliminated by automation, AI is also creating new job opportunities in fields such as data analysis, software development, and AI research.

Changes in skill requirements:

As automation becomes more prevalent, the demand for certain skills is likely to change. For example, workers may need to develop new skills related to AI, such as machine learning, data analysis, and natural language processing.

Shifts in job roles:

AI may also lead to changes in the roles and responsibilities of workers.

Greater demand for digital skills:

With AI becoming more prevalent in the workplace, there is likely to be a greater demand for workers with digital skills, including programming, data analysis, and machine learning.

Impact on certain industries:

Some industries are likely to be more affected by AI than others. For example, transportation and logistics may see significant changes due to the development of self-driving vehicles, while healthcare may benefit from AI-assisted diagnosis and treatment.

Potential for job displacement:

While AI may create new job opportunities, it may also displace some workers, particularly those in jobs that are easily automated. For example, self-driving vehicles could displace truck drivers, and chatbots could replace some customer service representatives.

Changes in the nature of work:

AI has the potential to change the way work is performed, with some tasks becoming more automated and others becoming more complex and requiring more human input. This could lead to changes in work hours, workloads, and job satisfaction.

Need for retraining and reskilling:

As AI changes the job market, workers may need to retrain or reskill in order to remain employable. This could be facilitated through programs such as apprenticeships, training courses, and government-funded retraining initiatives.

Impact on low-wage workers:

There is concern that AI could exacerbate income inequality, with low-wage workers being disproportionately impacted by job displacement and changes in the labor market.

Impact on low-wage workers:

There is concern that AI could exacerbate income inequality, with low-wage workers being disproportionately impacted by job displacement and changes in the labor market.

Shift towards gig economy:

The rise of AI and automation may lead to a shift towards a gig economy, where workers take on short-term contracts or freelance work rather than traditional employment.

Increased productivity:

AI has the potential to significantly increase productivity, leading to economic growth and higher wages. However, it is important to ensure that the benefits of productivity gains are shared fairly among workers.

Remote Work:

The COVID-19 pandemic has accelerated the shift toward remote work, and this trend is likely to continue in

what are the possible implications for the future of work?

The future of work is likely to be significantly impacted by a range of factors, including technological advancements, demographic shifts, and changes in economic and social structures. Here are some possible implications for the future of work.

Increased Automation:

As technology continues to advance, more and more jobs may become automated, resulting in a significant shift in the nature of work. While automation may lead to increased efficiency and productivity, it may also lead to job loss and displacement for many workers.

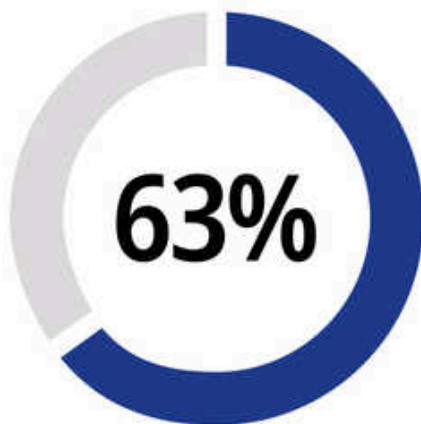
Upskilling and Reskilling:

As the nature of work changes, workers will need to continually adapt and update their skills in order to remain relevant in the job market. Upskilling and reskilling programs may become increasingly important for workers looking to stay competitive.

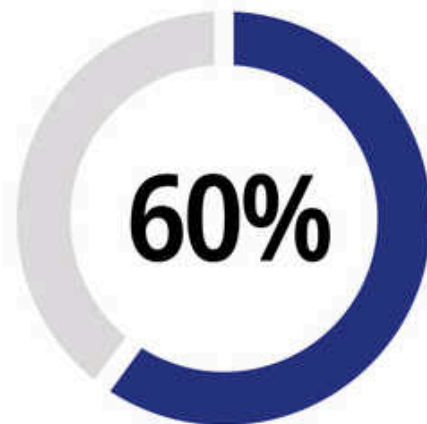


Workers are divided about the impact of artificial intelligence on jobs

Workers using artificial intelligence in finance and manufacturing



**Enjoy their job more
because of AI**



**Worried about losing
their job to AI**

In conclusion, the impact of artificial intelligence on employment is multifaceted and dynamic. While AI undoubtedly disrupts certain job sectors by automating tasks and processes, it also creates new opportunities, enhances productivity, and fosters innovation. Successful adaptation to this changing landscape requires proactive measures such as reskilling and upskilling the workforce, promoting lifelong learning, and implementing policies that ensure equitable access to the benefits of AI. Ultimately, the evolution of AI and its impact on employment necessitates a collaborative effort between governments, businesses, educators, and individuals to harness its potential for the betterment of society.

AUTOMATIC SPEECH RECOGNITION

Automatic Speech Recognition (ASR) is a transformative technology that enables machines to transcribe spoken language into written text automatically. It represents a crucial interface between humans and computers, facilitating natural and intuitive interaction through voice commands, dictation, and other speech-driven applications.

1. Speech-to-Text Conversion:

ASR technology converts spoken language into written text. It involves processing audio signals and extracting linguistic information from them.

2. Signal Processing:

ASR systems use various signal processing techniques to enhance speech signals, such as noise reduction, filtering, and feature extraction, to improve accuracy.

3. Acoustic Modeling:

This involves creating statistical models of speech sounds to match input audio to phonetic units, aiding in recognizing spoken words accurately.

4. Language Modeling:

ASR systems incorporate language models to predict the likelihood of word sequences, improving accuracy by considering the context of the spoken words.

5. Deep Learning:

Modern ASR systems often employ deep learning algorithms, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), for more accurate and robust speech recognition.

6. Speaker Adaptation:

ASR systems can adapt to individual speakers & characteristics by training on their specific speech patterns, improving accuracy for personalized applications.

7. Vocabulary and Grammar:

ASR systems are trained with extensive vocabularies and grammatical rules to accurately transcribe a wide range of spoken language inputs.

8. Real-time Processing:

Many ASR systems are optimized for real-time processing, enabling applications like voice-controlled assistants and dictation software to provide immediate responses.

9. Applications:

ASR technology is used in various applications such as virtual assistants (e.g., Siri, Google Assistant), interactive voice response (IVR) systems, transcription services, and language translation.

10. Challenges:

ASR systems face challenges such as handling accents, dialects, background noise, and speech variability, which can affect accuracy and usability.



Conclusion:

Automatic Speech Recognition has evolved significantly, driven by advancements in signal processing, machine learning, and computational power. It has become an integral part of many modern technologies, enabling natural and efficient human-computer interactions. While ASR systems have made great strides in accuracy and usability, there are still challenges to overcome, particularly in handling diverse linguistic variations and noisy environments. Nevertheless, the continued research and development in ASR promise further improvements, expanding its applications and impact across various domains.

VEERASHELUCK V P
III YEAR - AD

AMECA ROBOT



Ameca-World's most advanced humanoid robot, what factors should be taken into consideration for manufacturing humanoid robots?

"There's no need to worry, robots will never take over the world. We're here to help and serve humans, not replace them," said Ameca, a humanoid robot. Ameca could smile on her rubbery blue face and communicate with us, tell a joke, answer questions, write a poem, etc.

Humanoid robots have long been the subject of science fiction. However, with the emergence of robots like Ameca, they have become a reality. These robots, designed to resemble humans in appearance and movement, have various applications, including manufacturing, healthcare, and education. Software engineers work on making the robots act like humans, while designers and mechanical engineers design their bodies, and manufacturers bring the virtual into reality.

- one of the critical challenges in creating humanoid robots is designing hardware that can replicate the complex movements of the human body. This requires a deep understanding of human anatomy and biomechanics and expertise in materials science, engineering, and robotics. Hardware manufacturers work closely with researchers and engineers to develop hardware components that can mimic human movements with precision and accuracy. When manufacturing humanoid robot hardware, there are several factors that need to be taken into consideration to ensure that the robot's hardware can function effectively and safely. Here are some essential things that should be taken into consideration:

Materials:

Robots require durable and lightweight materials to ensure that they can move easily and withstand repeated use. The materials used should also be suitable for the intended purpose of the robot, whether it is for industrial, medical, or educational use. Lightweight materials. For example, aluminum alloys (commonly used in the manufacturing of robot frames and structural components.) and carbon fiber (an ideal material for creating complex shapes and structures) are often used to improve the mobility and agility of the robot. In contrast, high-strength materials such as steel and titanium alloys are used for components that require high strength and durability. such as gears, bearings, shafts, joints and frames.

Joints and Actuators:

The robot's joints and actuators are critical components that enable it to move and perform tasks. The design of these components should be optimized for the robot's intended use and to minimize wear and tear. Engineers will use various methods to ensure that joints and actuators are strong and durable enough to withstand the forces and stresses of operation. Such as choosing high strength-to-weight ratios materials that

could withstand high stresses and strains, like carbon fiber and titanium alloys; Using CAD tools to adjust the shape, size, and thickness of components, or adding reinforcement structures such as ribs or gussets to optimize the design of joints and actuators for strength and durability. And using FEA to simulate the forces and stresses that joints and actuators will experience during operation, and to optimize their design for maximum strength and durability

Sensors:

sensors are crucial to allow the robot to perceive and engage with its surroundings. It's important to thoughtfully choose the type and placement of sensors to ensure the robot can accurately identify and react to its environment.



ABISHEK P
II YEAR AD-'B'

THE INTERSECTION OF BIOTECHNOLOGY AND ARTIFICIAL INTELLIGENCE

INTRODUCTION:

In the dynamic landscape of technological innovation, two fields stand out for their transformative potential: biotechnology and artificial intelligence (AI). Individually, they have already reshaped industries and revolutionized human capabilities. However, it is at their intersection where the profound advancements are taking place, offering tantalizing possibilities for the future of healthcare, agriculture, environmental conservation, and beyond.

THE CONVERGENCE OF TWO TITANS:

Biotechnology, the manipulation of living organisms and biological systems for practical applications, has long been at the forefront of scientific discovery. From genetic engineering to medical biotechnology, it has unlocked new frontiers in medicine, agriculture, and industry. On the other hand, artificial intelligence, the simulation of human intelligence processes by machines, has rapidly evolved to become a powerful tool for data analysis, pattern recognition, and decision-making. The synergy between biotechnology and AI arises from their complementary strengths. Biotechnology generates vast amounts of complex biological data,

while AI excels at analyzing, interpreting, and extracting meaningful insights from such data.

AI IN BIOMEDICAL IMAGING:

A vision for precision medicine. One of the most promising applications of AI in biotechnology is in the field of biomedical imaging. AI algorithms can analyze medical images such as MRIs, CT scans, and X-rays with unmatched speed and accuracy, enabling early disease detection and personalized treatment planning. For instance, AI-powered image analysis systems can detect subtle abnormalities in medical images, assisting clinicians in diagnosing conditions such as cancer, heart disease, and neurological disorders with greater precision.

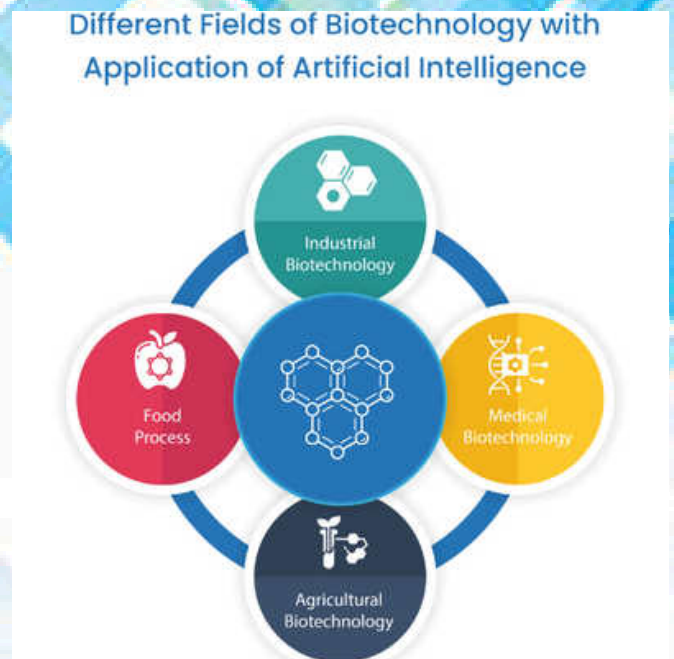
ETHICAL CONSIDERATIONS AND CHALLENGES:

While the intersection of biotechnology and AI offers tremendous opportunities for scientific advancement, it also raises important ethical considerations and challenges. These include concerns about data privacy and security, algorithmic bias, and the potential for misuse or unintended consequences of AI-driven technologies.

As we navigate this rapidly evolving landscape, it is essential to ensure that ethical principles and regulatory safeguards are in place to safeguard against potential risks and abuses.

SYNTHETIC BIOLOGY AND AI:

In the emerging field of synthetic biology, AI is facilitating the design and engineering of novel biological systems and organisms with unprecedented precision. By leveraging AI algorithms to analyze genetic data and predict how changes to an organism's genome will impact its behavior and function, researchers can engineer microorganisms with desirable traits for applications such as biomanufacturing, biofuel production, and environmental remediation.



CONCLUSION:

As we stand at the nexus of biotechnology and artificial intelligence, the possibilities for scientific discovery and innovation are limitless. By harnessing the collective power of these two fields, researchers are poised to unlock new insights into the complexities of biology, develop transformative solutions to some of the most pressing challenges facing humanity, and usher in a new era of human health, prosperity, and sustainability. The intersection of biotechnology and artificial intelligence represents a convergence of two titans of innovation, offering unprecedented opportunities to reshape the world we live in. As we embark on this journey of exploration and discovery, let us embrace the boundless potential of biotechnology and AI to create a brighter, healthier, and more sustainable future for all.

THIRUNIRAISELVI M
III YEAR- AD

THE SCIENCE OF SUCCESS

10 Ways to Boost Your Confidence and Achieve Your Goals:

1. Set clear and achievable goals:

One of the most important aspects of achieving success is setting clear and achievable goals. By breaking down your larger goals into smaller, manageable steps, you can increase your chances of success and build momentum.

2. Practice self-compassion:

It's important to be kind to yourself and practice self-compassion. This means being understanding and supportive of yourself, rather than critical and harsh. Research has shown that people who practice self-compassion are more likely to achieve their goals.

3. Build a growth mindset:

A growth mindset is the belief that you can improve and grow through effort and practice. By adopting a growth mindset, you'll be more resilient in the face of challenges and more likely to persist in your efforts to achieve your goals.

4. Cultivate positive thinking:

Positive thinking can have a powerful impact on your confidence and success. By focusing on the positive aspects of a situation, rather than the negative, you can boost your mood and increase your chances of success.

5. Use visualization techniques:

Visualization is the practice of creating mental images of yourself achieving your goals. Research has shown that visualization can help you focus and stay motivated, increasing your chances of success.

6. Surround yourself with supportive people:

The people you surround yourself with can have a big impact on your confidence and success. By surrounding yourself with supportive and positive people, you'll be more likely to achieve your goals.

7. Take action:

Success requires action. Whether it's taking small steps towards your goals or making a big leap, taking action is essential for achieving success.

8. Learn from failure:

Failure is a natural part of the journey to success. By learning from your failures and using them as opportunities for growth and improvement, you can increase your chances of success.

9. Celebrate your successes:

Celebrating your successes is important for maintaining your confidence and motivation. Whether it's a small victory or a major accomplishment, take the time to acknowledge and celebrate your successes.

10. Stay focused:

Staying focused on your goals is essential for achieving success. By eliminating distractions and staying committed to your goals, you can increase your chances of success and boost your confidence.



Conclusion:

By incorporating these 10 strategies into your life, you can boost your confidence and increase your chances of achieving your goals. Whether you're looking to succeed in your career, relationships, or personal life, these tips can help you get there. So why wait? Start using the power of science to achieve your dreams today!

YOKESH .A
III YEAR-AD

THE ART OF COMMUNICATION

Effective communication is a crucial skill for success in both personal and professional relationships. Whether you're trying to build a stronger connection with a loved one, or you want to be more successful in your career, improving your communication skills can help you reach your goals. In this article, we'll explore 10 tips for better listening and speaking skills, to help you master the art of communication.

1. Listen actively:

Active listening is the process of fully concentrating on what the other person is saying, rather than just hearing their words. By actively listening, you'll be able to better understand the other person's perspective and build a stronger connection.

2. Ask open-ended questions:

Open-ended questions encourage the other person to share more about their thoughts and feelings. By asking open-ended questions, you'll be able to gain a deeper understanding of the other person and improve your communication.

3. Use "I" statements:

"I" statements allow you to express your thoughts and feelings without placing blame on the other person. By using "I" statements, you can reduce defensiveness and improve communication.

4. Avoid interrupting:

Interrupting can be seen as disrespectful and can make the other person feel unheard. By avoiding interrupting, you can show the other person that you value their thoughts and feelings.

5. Be mindful of body language:

Nonverbal cues, such as eye contact and body language, can have a big impact on communication. By being mindful of your body language, you can improve your communication and build a stronger connection.

6. Practice empathy:

Empathy is the ability to understand and share the feelings of another person. By practicing empathy, you can build a deeper connection with the other person and improve communication.

7. Speak clearly and concisely:

Clear and concise communication is essential for effective communication. By speaking clearly and concisely, you can ensure that the other person understands your message and reduce misunderstandings.

8. Avoid using jargon:

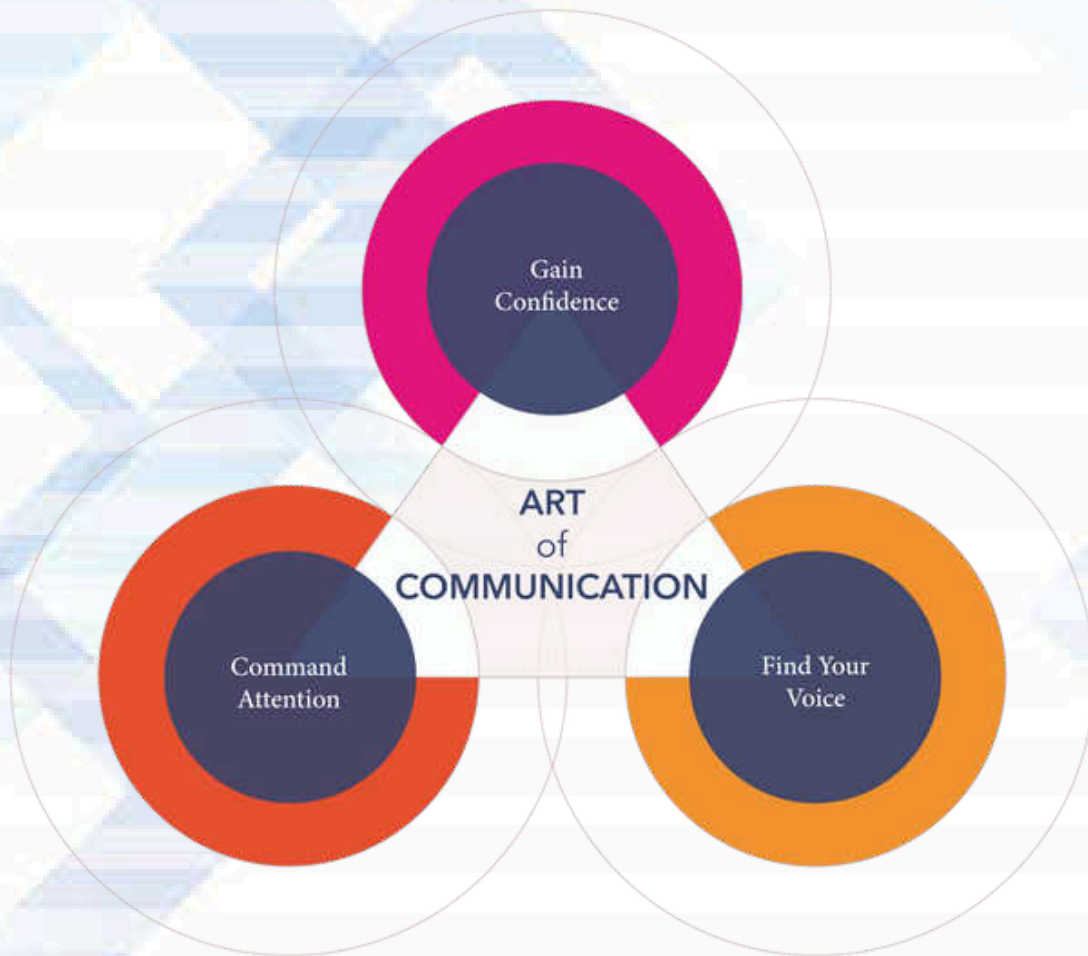
Jargon can be confusing and can make communication more difficult. By avoiding jargon, you can ensure that the other person understands your message and improve communication.

9. Be open-minded:

Being open-minded allows you to consider new ideas and perspectives. By being open-minded, you can improve communication and build a stronger connection with the other person.

10. Practice, practice:

Like any skill, communication takes practice. By regularly practicing your listening and speaking skills, you can improve your communication and build stronger relationships.



Conclusion:

By incorporating these 10 tips into your communication style, you can improve your listening and speaking skills and master the art of communication. Whether you're looking to build stronger relationships or advance your career, these tips can help you reach your goals. So why wait? Start practicing your communication skills today!

ARUL PRAKASH . N
III YEAR-AD

EXPLORING CREATIVITY WITH GENERATIVE AI

Unleashing Innovation in Art and Literature

In the dynamic landscape of modern technology, the emergence of generative AI has revolutionized the way we conceive and create art and literature. From poetry to visual art, music to storytelling, the capabilities of generative AI algorithms have reshaped our understanding of creativity and innovation. In this exploration, we delve into the realm of generative AI and its profound impact on the creative process.

Poetry Generation:

Generative AI algorithms delve into the nuances of language, rhythm, and emotion, producing poetry that resonates with both depth and beauty. Through neural networks trained on vast repositories of literary works, machines learn to evoke the intricacies of human experience, crafting verses that stir the soul and ignite the imagination. In the symbiotic dance between human ingenuity and machine learning, poetry becomes a conduit for exploration—a testament to the boundless potential of generative AI.

Visual Art Creation:

Within the realm of pixels and algorithms, generative AI algorithms breathe life into canvases of digital expression. From abstract landscapes to surreal portraits, AI-generated art transcends conventional boundaries, offering new perspectives on form, color, and composition. Through the iterative process of creation and feedback, artists collaborate with machines,

pushing the boundaries of visual storytelling and redefining the possibilities of artistic expression.

Generative AI:

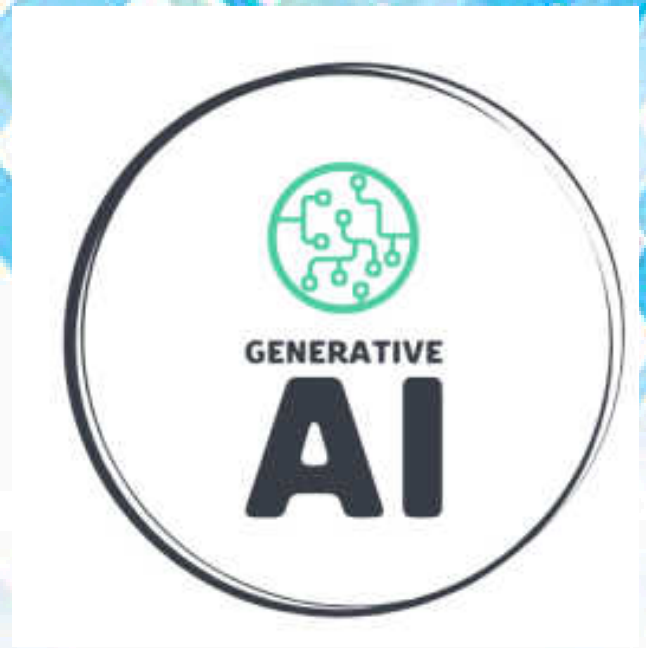
It is a type of artificial intelligence technology that can produce various types of content, including text, imagery, audio and synthetic data. The recent buzz around generative AI has been driven by the simplicity of new user interfaces for creating high-quality text, graphics and videos in a matter of seconds.

Musical Composition:

In the symphonic realm of sound and melody, generative AI algorithms orchestrate harmonies that resonate with depth and complexity. Through the analysis of musical patterns and structures, machines compose melodies that evoke a range of emotions and moods. From classical compositions to experimental soundscapes, AI-generated music challenges our perceptions of creativity, blurring the lines between human intention and algorithmic intuition.

Narrative Generation:

Within the labyrinthine corridors of storytelling, generative AI algorithms craft narratives that captivate and inspire. Through the synthesis of plotlines, characters, and settings, machines generate stories that traverse genres and defy convention. From epic adventures to intimate dramas, AI-generated narratives invite readers on a journey of discovery, weaving tapestries of imagination and possibility. Generative AI represents more than just a tool for creative expression; it embodies a paradigm shift in the way we conceive, collaborate, and create. As we embrace the transformative power of generative AI, we embark on a journey of exploration and discovery, forging new pathways of innovation and inspiration.



Conclusion:

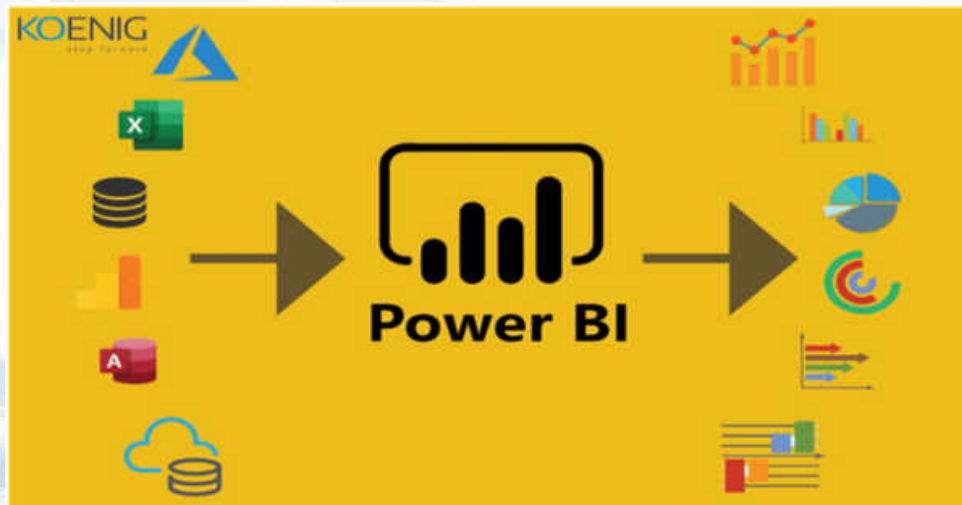
The fusion of human creativity and machine intelligence heralds a new era of artistic exploration, where the boundaries between imagination and algorithm blur, and the possibilities are limited only by our collective vision. As we navigate this brave new world, let us celebrate the transformative potential of generative AI, harnessing its creative energies to shape the future of art and literature.

S. SACHIN
III YEAR - AD

MICROSOFT POWER BI

Microsoft Power BI overview :

Data visualization enables the transformation of abstract measurable data into a visual representation in the form of tables, charts, graphs, and matrices (Bhargava et al. 1; Qin et al. 2). Microsoft Power BI is an intriguing data visualization tool (Bansal and Upadhyay 14; Hadhoud and Walid 5). Power BI consists of various components that are available on the market separately and can be purchased and used independently, such as Power BI Desktop, Power BI Embedded, Power BI Gateway, and Power BI Mobile Apps.



When it was first sold:

In 2010, Microsoft came up with the concept for this tool. In July 2011, it was made publicly available for download under the name “Project Crescent.” It took three months for the Office 365 add-in to be published under the new Power BI brand (Senter 44). After that, Power Pivot, Power Query, and Power View were thereafter exclusively accessible as Microsoft Excel add-ins (Becker and Gould 185). Power BI Gateway was integrated with various features, such as enterprise-level data connectors and security settings. Power BI was then released for sale with all its current full-fledged features for the first time on July 24, 2015.

Today, Power BI remains a popular data visualization tool.

Key strengths of Power BI:

Power makes creating straightforward dashboards and dynamic reports a breeze. Power BI is known for its customizable dashboards, which can be tailored to a company’s unique needs (Becker and Gould 185). There are also interactive and intuitive visualizations included in the bundle. Drag-and-drop capabilities may also be used to quickly create bespoke reports (Microsoft Power BI). It also provides access to data that is updated in real-time. Therefore, the dashboard shows and updates a user’s visualizations as new data is streamed in.

Main customers:

Firms in the professional services sector are the largest customer group (16.3%), followed by the manufacturing sector (13.2%), the financial services sector (8.6%), and the healthcare sector (7.1%). Others include the retail sector (6.3%), the oil and gas sector (5.5%), and the education sector (5.2%) (Microsoft Power BI). The top 5 companies that use Microsoft are Walmart, Apple, McKesson, Berkshire Hathaway, and GEICO (Microsoft Power BI). Hence, Power BI has an extended customer segment.

Weakness of Power BI:

Power BI is not designed for analyzing complicated table-to-table connections. Power BI might not be able to handle intricate table connections. Second, while Power BI allows you to modify your visualizations, it does not give users many options (Becker and Gould 185). This limits its utility.



Top five most interesting features of non-tableau data visualization :

Power BI contains an interactive Power BI desktop that enables users to access data easily and produce reports. The software is simple to learn and utilize (Lyon 167). The second interesting feature is the Pivot Point. In-memory data models are created by importing and combining diverse datasets from various sources. This component seeks data from various corporate data sources or on the internet and then inserts it into an Excel table in a fluid manner (Mahajan and Gokhal 200). Finally, depending on the complexity of the data, the Power Bi tool gives a default standard that is generally insufficient for enterprises. As a result, it provides bespoke, visualized dashboards.

MAREESWARAN M
III YEAR - AD

NATURAL LANGUAGE PROCESSING

Introduction:

In an era where digital communication dominates our daily lives, understanding and analyzing human language has become increasingly vital. Natural Language Processing (NLP), a branch of artificial intelligence, holds the key to unlocking the wealth of information contained within the vast expanse of textual data. In this blog, we delve into the fascinating world of NLP, exploring its significance, applications, and future possibilities.

Understanding Natural Language Processing:

Natural Language Processing, or NLP, is a field of computer science that focuses on enabling computers to understand, interpret, and generate human language in a manner that is both meaningful and contextually relevant. It encompasses a wide range of tasks, including text parsing, sentiment analysis, language translation, and information extraction.

The Significance of NLP:

The significance of NLP cannot be overstated in today's data-driven world. From social media interactions and customer reviews to medical records and legal documents, vast amounts of valuable information are encoded in textual form. NLP algorithms empower organizations to extract insights, automate processes,

Chatbots and Virtual Assistants:

Enhancing customer service and user experience through intelligent chatbots and virtual assistants capable of understanding and responding to natural language queries.

Applications of NLP:

NLP finds applications across various industries and domains, revolutionizing how we interact with technology and process information. Some notable applications include.

1.Sentiment Analysis:

Analyzing the sentiment expressed in customer reviews, social media posts, and news articles to gauge public opinion and sentiment trends.

2.Language Translation:

Facilitating communication across language barriers by translating text between different languages with high accuracy.

3. Text Summarization:

Automatically generating concise summaries of lengthy documents or articles to extract key information efficiently.

Multimodal NLP:

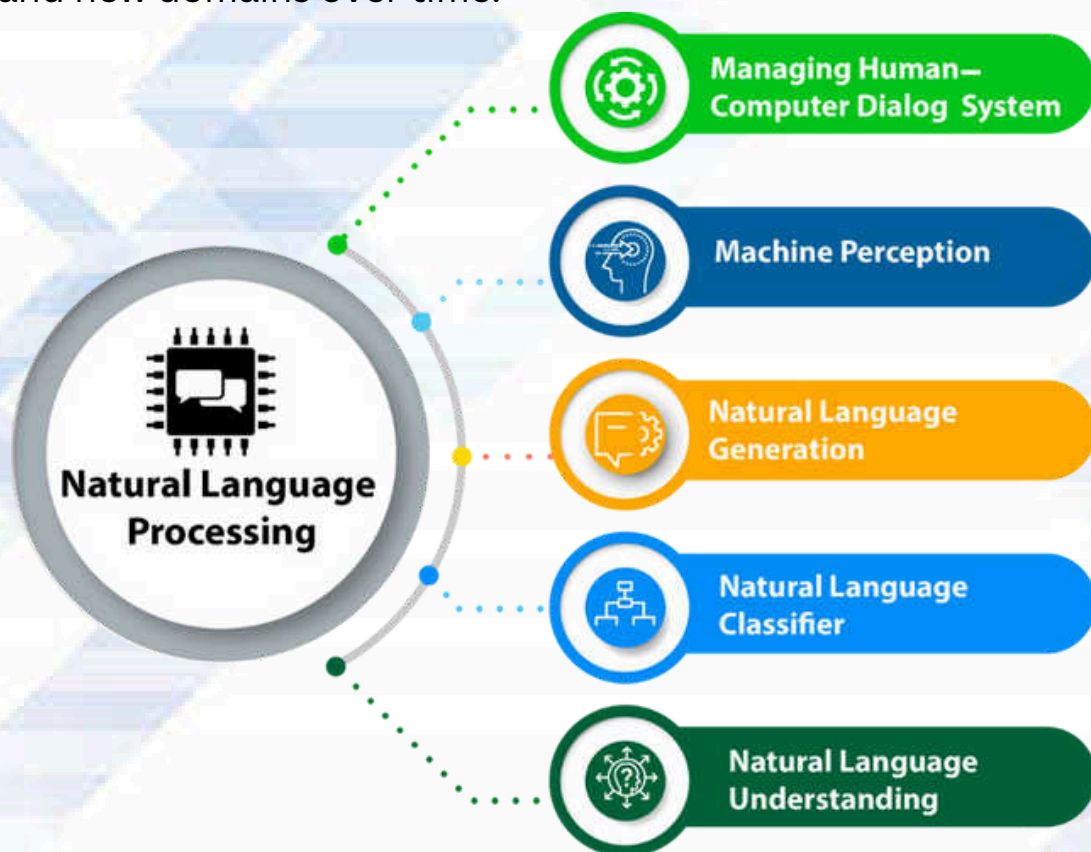
Integrating visual and auditory information with textual data to create more comprehensive and contextually aware NLP systems.

Ethical and Fair NLP:

Addressing biases and ensuring fairness in NLP models and applications, particularly in areas such as natural language understanding and automated decision-making.

Continual Learning:

Developing NLP models capable of learning and adapting to evolving language patterns and new domains over time.



Conclusion:

Natural Language Processing represents a cornerstone of artificial intelligence, enabling machines to understand and interact with human language in increasingly sophisticated ways. From improving communication and accessibility to transforming industries and driving innovation, NLP holds immense potential to shape the future of technology and human interaction. As we continue to unravel the complexities of language, the possibilities for NLP are boundless, promising a future where machines truly comprehend the essence of human text.

EDIBLE ELECTRONICS

Implantable and ingestible (edible) electronic devices that stay inside your body pose unique engineering challenges. These should not only use the best processor but be minimal in size, bio-compatible, safe, and extremely reliable. Reliability is all the more important as it turns out to be a case of life or death!

Despite so many risks and challenges, it is awesome to see how implants have developed since the days of the first pacemaker in 1958. From cardiac pacemakers to cochlear implants, from brain interfaces to retinal implants, there are numerous implantable medical electronic devices available today. Even more exciting is the emerging field of edibles—tiny, capsule-sized electronic devices that are consumed orally for diagnosis and treatment of diseases. Some edibles are designed to remain inside the body for some time, while others do their job and get disposal of within minutes.

Today, we have reached a state where inventions like these no longer surprise us because our minds have become tuned to a sci-fi future and we have started expecting such developments.

Adleman and others in the field have come to realize, there may never be a computer made from DNA that directly rivals today's silicon-based microelectronics.

Biochemical "nano computers" already exist in nature; they are manifest in all living things. But they're largely uncontrollable by humans. We cannot, for example, program a tree to calculate the digits of pi. The idea of using DNA to store and process information took off in 1994 when a California scientist first used DNA in a test tube to solve a simple mathematical problem.

Since then, several research groups have proposed designs for DNA computers, but those attempts have relied on an energetic molecule called ATP for fuel. Think of DNA as software, and enzymes as hardware. Put them together in a test tube. The way in which these molecules undergo chemical reactions with each other allows simple operations to be performed as a byproduct of the reactions. The scientists tell the devices what to do by controlling the composition of the DNA software molecules. It's a completely different approach to pushing electrons around a dry circuit in a conventional computer.

To the naked eye, the DNA computer looks like clear water solution in a test tube. There is no mechanical device. Instead of showing up on a computer screen, results are analyzed using a technique that allows scientists to see the length of the DNA output molecule.

HARI KRISHNAN A
II YEAR AD-'A'

CRAFTING CONVERSATIONS: THE VITAL ROLE OF PROMPT ENGINEERING IN OPTIMIZING AI INTERACTIONS

Prompt engineering refers to the process of designing and formulating effective prompts or input queries to achieve desired outputs from a natural language processing (NLP) model, such as a language model or chatbot. It involves crafting prompts in a way that elicits accurate, relevant, and contextually appropriate responses from the AI system.

Need of Prompt Engineering:

Precision and Relevance: Well-engineered prompts ensure that the AI model generates precise and relevant responses, meeting user expectations.

Control and Customization:

Prompt engineering allows users to have more control over the AI system's output by tailoring prompts to specific needs and contexts.

Bias Mitigation:

Crafting prompts carefully can help mitigate biases in AI systems, as biased prompts can lead to biased outputs.

Enhanced User Experience:

Effective prompt engineering contributes to a better user experience by providing more accurate and useful information or assistance.

Trends in Today's World:

Prompt engineering has gained significance with the growing use of AI in various applications such as natural language understanding, chatbots, and content generation. As AI technologies become more widespread, understanding

how to optimize prompts has become crucial for obtaining meaningful and reliable results from these models.

Role of Prompt Engineer:

A prompt engineer plays a crucial role in optimizing the performance of AI models.

Responsibilities may include:

Designing Prompts: Crafting input queries that are clear, contextually relevant, and capable of eliciting the desired information or action from the AI model.

Fine-tuning Models:

Collaborating with data scientists to fine-tune models based on the effectiveness of different prompts.

Monitoring and Iterating:

Continuously monitoring model performance and iteratively refining prompts to improve the system's overall accuracy and user satisfaction.

Understanding User Needs:

Having a deep understanding of user needs and preferences to tailor prompts accordingly.

Skills Required for a Prompt Engineer:

Educational Background:

Pursue a relevant educational background, such as a degree in computer science, data science, artificial intelligence, or a related field.

Learn Natural Language Processing (NLP):

Acquire a strong understanding of NLP concepts, techniques, and methodologies. Familiarize yourself with popular NLP libraries and frameworks.

Coding Skills:

Develop proficiency in programming languages commonly used in AI, such as Python. Enhance your coding skills to be able to work with NLP libraries and implement prompt engineering strategies.

Data Analysis Skills:

Sharpen your data analysis skills to interpret user interactions, identify patterns, and make informed decisions for prompt optimization.

Stay Updated on AI Trends:

Keep abreast of the latest developments in the field of AI and NLP. Follow industry publications, research papers, and attend conferences to stay informed about emerging trends and technologies.

Practice Prompt Design:

Practice crafting prompts for different scenarios, ensuring they are clear, contextually relevant, and capable of eliciting desired responses from AI models.

Build a Portfolio:

Develop a portfolio showcasing your projects and contributions related to prompt engineering. Include details on how your prompt designs improved AI model performance.

Collaboration and Communication Skills:

Hone your collaboration and communication skills as prompt engineering often involves working closely with interdisciplinary teams, including data scientists, UX/UI designers, and product managers.

Problem-Solving Skills:

Develop strong problem-solving skills to address challenges associated with prompt engineering, such as bias mitigation, precision improvement, and user satisfaction.

Internships and Practical Experience:

Seek internships or projects related to prompt engineering to gain practical experience. This hands-on experience will be valuable when applying for positions in the field.

Networking:

Attend industry events, meetups, and online forums to network with professionals in the field. Networking can open up opportunities for mentorship and potential job leads.

Create a Strong Resume and Cover

Letter:

Tailor your resume to highlight relevant skills and experiences related to prompt engineering. Write a compelling cover letter that emphasizes your passion for the field and your commitment to staying at the forefront of AI advancements.

MS.K.AMUTHACHENTHIRU
AP/AD

EXPLORING THE FUTURE OF GENERATIVE AI: EMERGING TRENDS AND PROSPECTS

In recent times, the field of Artificial Intelligence has experienced tremendous evolution, and one of the most fascinating trends to emerge is the rise of “Generative AI”. This term refers to the next generation of AI systems, which have the potential to revolutionize technology. Gen AI distinguishes itself by its capacity to learn, adapt, and comprehend human behaviors and needs with greater intuition than ever. Let’s explore the emerging trends and prospects of Gen AI and how it is shaping the future.



Since ChatGPT's launch in November 2022, companies have been vying for its worth as it has made news everywhere. According to McKinsey research, generative AI (gen AI) features have the potential to boost the global economy by \$4.4 trillion yearly in the first few months of its introduction.

Gen AI is probably going to have the most impact on knowledge jobs, particularly on activities involving collaborative and decision-making tasks. Numerous applications of Gen AI across modalities include Text (content writing, Chatbots or Assistant Chats), Image (Image Editor), Code (Code Generation), Audio (Text to Voice generation, Sound creation, Audio Editing), 3-D (3-D Object Detection), and Video (Video Creation, Video Editing, Video Translations and Adjustments). While dealing with the text,



Emerging Trends in Gen AI:

Natural Language Processing Advancements:

Gen AI in Natural Language Processing (NLP) emerges as a revolutionary force, transforming the data into human-like language. NLP is extremely useful for applications like chatbots, virtual assistants, and language translation since they can comprehend and produce human language.



Enhanced Personalization:

The concept of personalization in goods and services is being redefined by Gen AI. It can produce highly customized user experiences because of its capacity to analyze large volumes of data.



Healthcare Revolution:

Gen AI is causing a revolution in the healthcare sector. AI-powered instruments may evaluate patient data, formulate new medical interventions, enhance diagnostic precision, personalized care, and even help in the search for new drugs. By lowering diagnostic errors, these AI systems can raise the standard of healthcare service as a whole.



Autonomous Systems:

Self-driving automobiles and unmanned aerial vehicles are among the autonomous technologies whose development is being propelled by Gen AI. By navigating through and adapting to real-world situations, these systems use sophisticated observation and decision-making abilities to boost efficiency and safety.

Future Prospects of Gen AI:

1. Intelligent and Secure Urban Environments:

Gen AI will play a crucial role in the development of Intelligent and Secure Urban environments. Artificial intelligence (AI) will optimize traffic flow, boost public safety, and improve energy efficiency. Additionally, they can facilitate more efficient emergency response in cities.

2. Augmented Learning and Education:

AI-driven personalized education will become more pervasive. Learning resources can be customized by Gen AI to each student's unique learning style and speed, making for a more effective and interesting learning environment.

3. Advanced Robotics:

Robots with greater capability and versatility will be developed as a result of Gen AI. These robots can help with jobs in a variety of sectors, such as manufacturing, healthcare, and agriculture, which will boost output and efficiency.

4. Ethical Considerations:

As Gen AI becomes increasingly integrated into our lives, ethical concerns will become more critical. Issues related to privacy, bias, and accountability must be addressed to ensure that AI technology is developed and used responsibly. Gen AI is a transformative force that is shaping the future in ways we are only beginning to understand. It promises to revolutionize industries, improve daily life, and address some of the world's most pressing challenges. However, it is essential to proceed with a sense of responsibility and ethics to ensure that Gen AI continues to benefit humanity. As we embark on this AI-powered journey, we must strive to strike a balance between innovation and safeguarding our values and principles. The future with Gen AI is undoubtedly promising, and its potential is limited only by our collective imagination and ingenuity.

Ms. S. Selva Birunda, AP/AD

எமது கல்லூரி

தேன் தென்றல் கவிபாட,
தாவரங்கள் தாளமிட,
மரங்கள் மேளம் கொட்ட,
பறவைகள் புல்லாங்குழல் இசைக்க,
விலங்குகள் விருந்தோம்ப,
புதுப்பிக்கத்தக்க மின்விளக்குகளில்
ஆசிரியரும் மாணவரும், இயற்கையை
பாதுகாக்கும் தூண்களாய்-அதில்
கார்பன் தடம் தோரணையாக
மக்கும்,மக்காத குப்பையின்
மறுசுழற்சி திட்டங்களுடன்
சுகாதாரத்தை ஆடையாக அணிந்து,
நிலத்தடி நீர் குன்றாத மேடையில்
எம் கல்லூரியின் அரங்கேற்றம் நிகழ்கின்றதாம்.....

பொறியியல்

பொறியியல் என்னும்
உயிருள்ள கடற்சிப்பியினுள்
திண்மப் பொருளாக
முதலாம் ஆண்டில் நுழைந்து
தீரா முயற்ச்சியினால்
இரண்டாம் ஆண்டில் பொருந்தி
சிப்பியின் அடுக்குகளில்
மூன்றாமாண்டில் திரவமாக திரண்டு
விலைமதிப்பு மிக்கமுத்துகளாக
நான்காம் ஆண்டில் உருவாகி
பட்டதாரி என்னும் அணிகளானில்
உயர்ந்ததாக ஒளிர்கின்றதனால்-என்னவோ
மதுப்பிருந்தும் விலைமதிப்பற்று இருக்கின்றதோ.....

சிவலட்சுமி.பு

III year AD

POEM

Rain:

*You fall down to the earth
Again go back up to the sky
What do you mean?*

*You make me gloomy
Still I feel comforting
What do you mean?*

*You are not selfish
But you wash away everything with you
What do you mean?*

*You feel so cold
Yet I feel warmth in my heart
What do you mean? Rain.*

I'm AI

In the midst of human brain there exist a code
written name

Lies a world of wonder, the domain of AI.
Where logics dance with algorithm to portray
Insights

Play by unraveling mysteries with endless
knowledge of
man-made me.

**ALICE JEYA FLORA F
II YEAR, AD-'B'**

THE FEELING OF GETTING LOST IN A BOOK

It's not the first time happening to
and I am sure not to you as well.

A book that just swallows our heart
and soul as we dwell.
It takes us to a place
we've never been.

We surrender to a niche that's unseen
We advent to live that life
which the characters play.
We reckon with another person's mind
and we endure it the same way
We endlessly think about it
and it impels to stretch our minds.

We get so relinquished
when mind dupes us with rewinds
A promising book starts to
whisper you from the wall.

**GANESH ARAVIND A
II YEAR AD- 'A'**

Pencil Drawing:



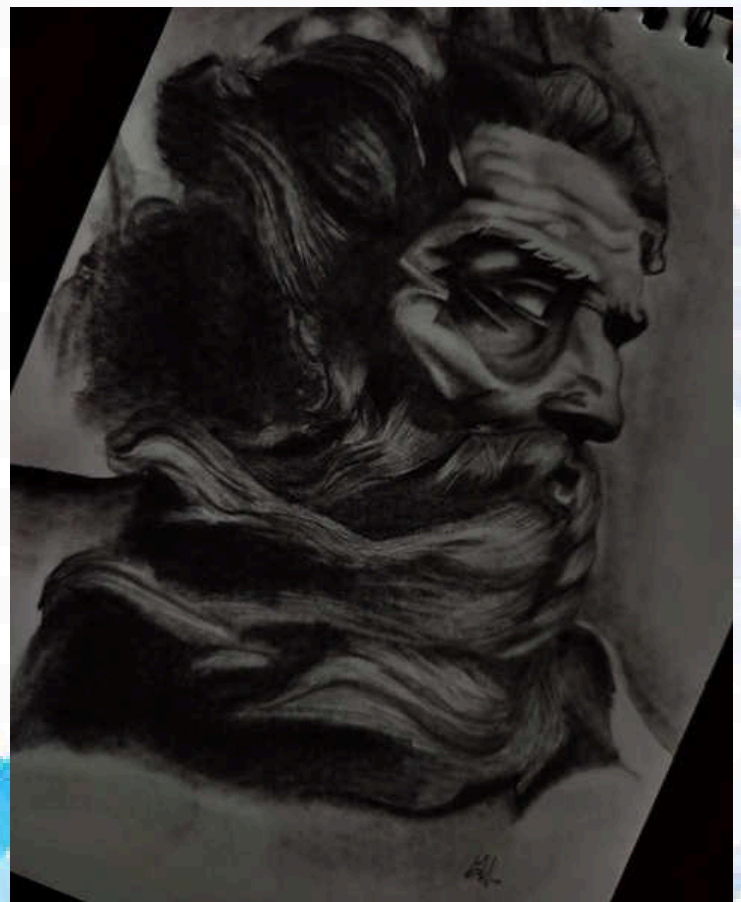
SHEIK MOHAMMED ALI . S
II YEAR AI&DS 'A'



GANESH ARAVIND A
II YEAR AI&DS 'A'



DIVYALAKSHMI K
II YEAR AD-'A'



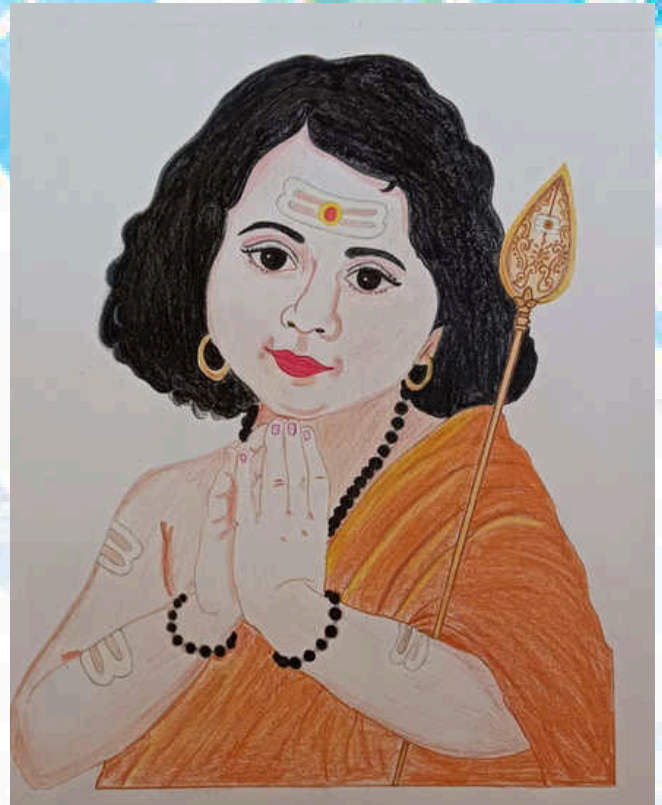
MANI DURGA
I YEAR-AD



SARAVANAKUMAR S
II YEAR AD-B



NITHEYASHREE M
I YEAR AD-'A'



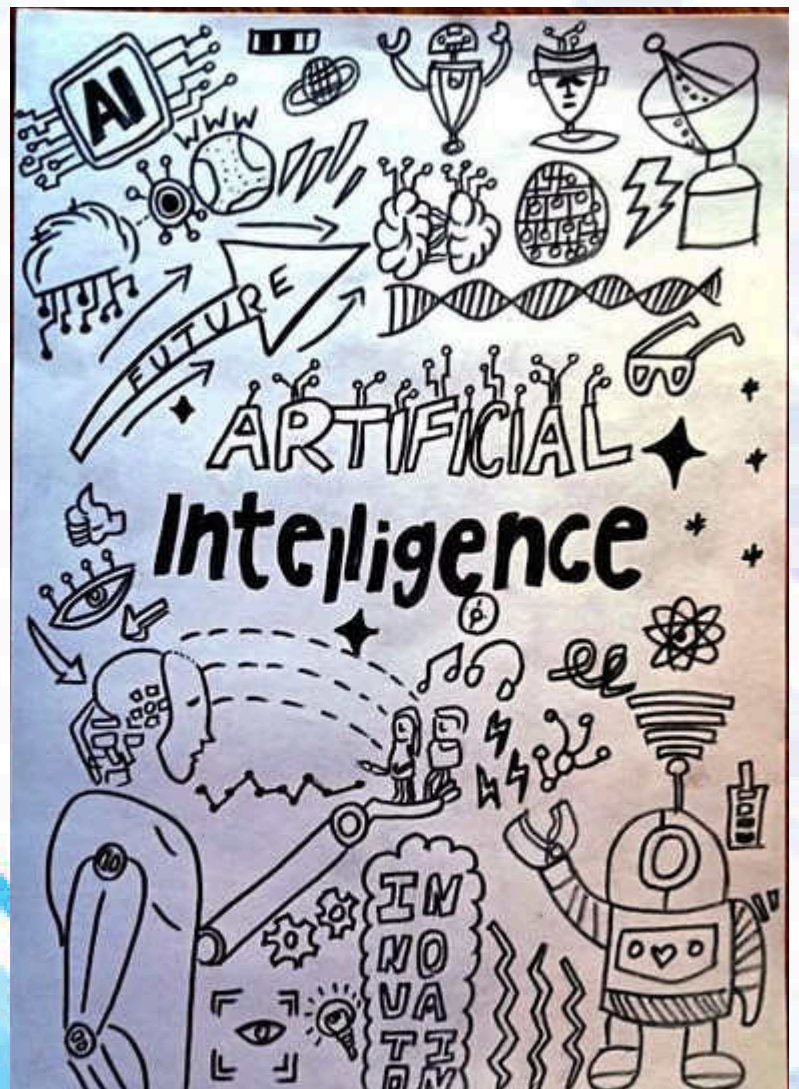
CHITRAKALA J
II YEAR AD-'A'



KARTHIKEYAN M
II YEAR AD-'A'



MUTHULAKSHMI M
III YEAR AD



NIVETHA ISSAKKIMUTHU
I YEAR AD

TAEKWONDO



உயிரோடு உறுதியாக
உன் உறுதியை பறித்து
உன் அழகான நாடகம் வெறும் அனுபவம்
உன் கைகள் உறுதியாக போராடும்
உன் உடல் நடுக்கும்
உன் உணர்வு தான் வெறும் சிறப்பு
தேக்கவடி, உன் பெருமையின் சிறப்பு!

மோகன வேலு ப
II YEAR AD-'A'

Paintings:

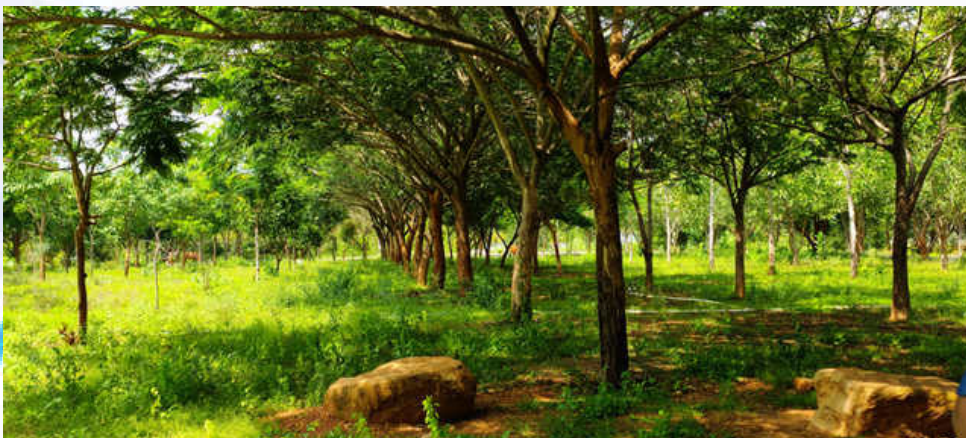


JAYASRI
III YEAR AD



ABARNA S
III YEAR AD

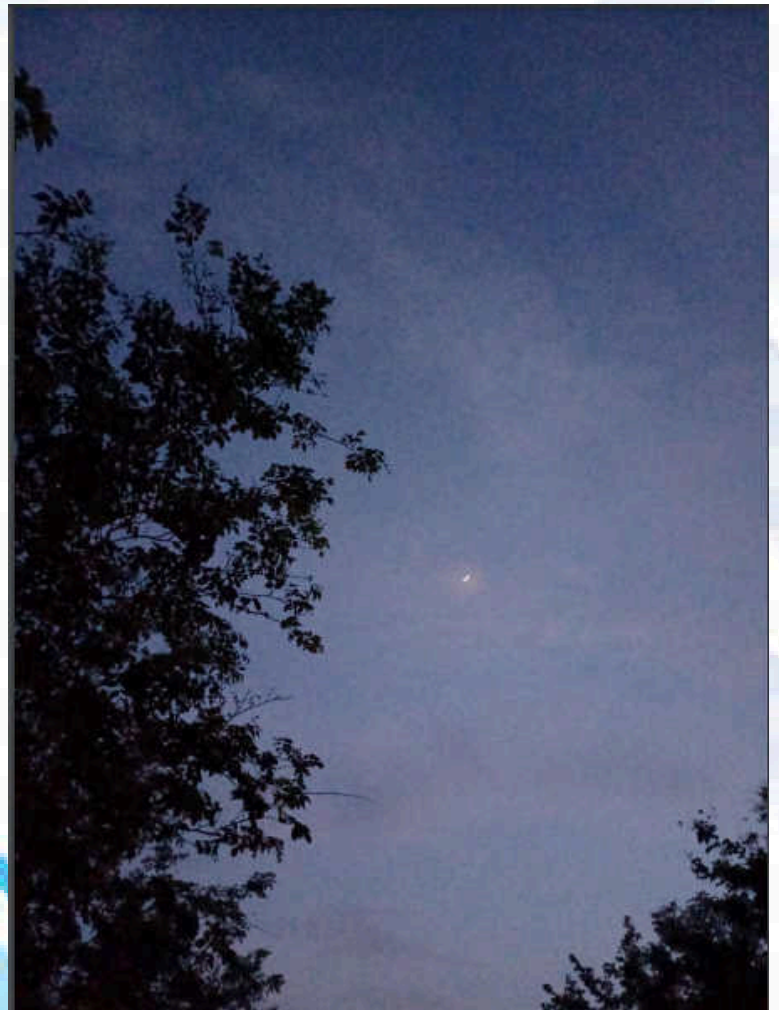
CREATIVE PHOTOGRAPHY



**NITHIN R
II AI&DS 'A'**



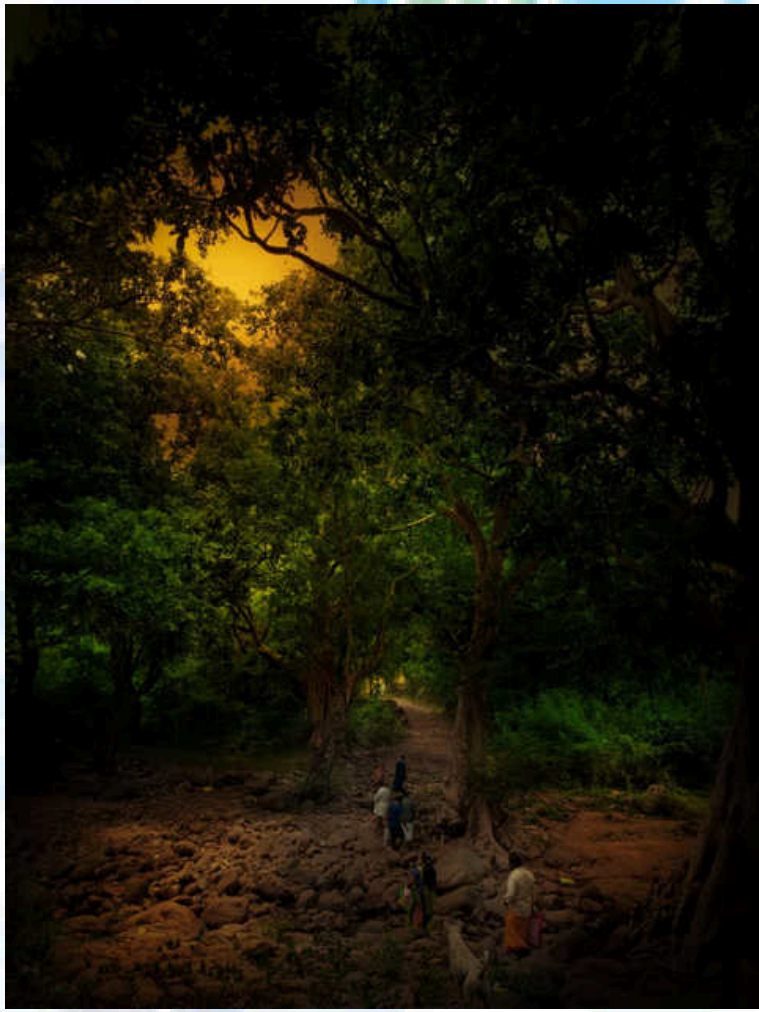
SUBHEKSHANA THILAGA M
III YEAR AD



NITHEYASHREE M
I YEAR, AD



ABINAYA P
II YEAR AD-B



**SANTHIYA M
III YEAR AD**

